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Motor disconnection detection while the motor is switched on

Abstract

A system for detecting motor disconnections while a motor is switched on comprises a motor, a motor drive circuit, and a processor. The processor determines whether a set of disconnection conditions is met for more than a threshold time. In this process, the processor determines that a pulse signal that leads to power the motor is enabled, that a speed associated with the motor is more than a threshold speed, a drive voltage output associated with the motor drive circuit is more than a threshold voltage, a current applied to the motor is less than a threshold current, and that the speed associated with the motor is zero. The processor determines that at least one terminal of the motor is disconnected from the motor drive circuit in response to determining that the set of disconnection conditions is met for more than the threshold time.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to heating, ventilation, and air conditioning (HVAC) systems and methods of their use, and more specifically to motor disconnection detection while the motor is switched on.

BACKGROUND

(2) Heating, ventilation, and air conditioning (HVAC) systems are used to regulate environmental conditions within an enclosed space. Typically, HVAC systems include an evaporator coil and a condenser coil. A blower of the HVAC system pulls warm air from the enclosed space and pushes the air across the evaporator coil to cool the air. The air is cooled via heat transfer with refrigerant flowing through the evaporator coil and returned to the enclosed space as conditioned air. Meanwhile, the refrigerant flowing through the evaporator is heated and generally transitions to the vapor phase. After being pressurized by a compressor, the refrigerant from the evaporator coil flows toward the condenser coil where it is cooled before flowing back to the evaporator coil to repeat the cycle. These components of the HVAC system are run by motors. Motors may be disconnected from one or more terminals. If disconnected, the motor and motor-driven components may become unstable.

SUMMARY

(3) The system described in the present application provides several practical applications and technical advantages that overcome the current technical problems described herein. The following disclosure is particularly integrated into a practical application of detecting motor disconnection at one or more motor terminals in cases where the motor is switched off and in cases where the motor is switched on. This, in turn, provides an additional practical application of improving the operation of the motor and other components that the motor is tasked to operate. For example, in a Heating, Ventilation, and Air Conditioning (HVAC) system, a motor may be tasked to power a fan, a blower, a compressor, or other components of the HVAC system. If the motor disconnection is left undetected, the motor-driven component (e.g., the fan, the blower, the compressor, etc.) may not

operate as intended and/or unstable or irregular voltage/current coming from the motor may damage the motor-driven component. By detecting the motor disconnection, the disconnection may be addressed by an operator. Thus, the operations of the motor and the motor-driven component are improved.

(4) In some cases, a terminal (e.g., any of the terminals a, b, c) of the motor may be disconnected due to hardware and/or software malfunctions. The current approaches do not provide a solution to detect motor disconnection. In other words, using the current approaches, it is not known whether the motor is disconnected, much less the location of the disconnection, which motor terminal or from which side of the motor—from the frontend or the backend.

(5) The disclosed system is configured to provide a technical solution to the technical problem discussed above and other technical problems arising in HVAC technology. For example, the disclosed system is configured to provide a method to detect disconnection at one or more terminals of the motor. In certain embodiments, the disclosed system is configured to detect the motor disconnection while the motor is switched off (e.g., before it is running). In certain embodiments, the disclosed system is configured to detect the motor disconnection while the motor is switched on.

(6) Motor Disconnection Detection while the Motor is Switched Off

(7) In some embodiments, a system for detecting motor disconnection at motor terminals before the motor is switched on comprises a motor, a motor drive circuit, and a processor. The motor is configured to operate a motor-driven component, wherein the motor has three terminals communicatively coupled to the motor-driven component. The motor drive circuit is configured to power the motor. The processor is communicatively coupled with the motor drive circuit. The processor is configured to determine whether a pulse signal that powers the motor is enabled. The processor is further configured to determine whether a permanent magnet associated with the motor is being aligned with at least one winding associated with the motor in response to determining that the pulse signal is enabled. The processor is further configured to communicate, to the motor drive circuit, a control signal that indicates to transmit a DC voltage signal to a first terminal from among the terminals at the motor in response to determining that the permanent magnet is being aligned with at least one winding. The processor is further configured to determine a first current value at the first terminal of the motor. The processor is further configured to compare the first current value with a first threshold current value. The processor is further configured to determine that the first current value is less than the first threshold current value. The processor is further configured to determine that the first terminal of the motor is disconnected from the motor drive circuit in response to determining that the first current value is less than the first threshold current value.

(8) Motor Disconnection Detection while the Motor is Switched on

(9) In some embodiments, a system for detecting motor disconnection at motor terminals while the motor is switched on comprises a motor, a motor drive circuit, and a processor. The motor is configured to operate a motor-driven component. The motor drive circuit is configured to power the motor. The processor is communicatively coupled to the motor drive circuit. The processor is configured to determine whether a set of disconnection conditions is met for more than a threshold time. The processor is further configured to determine that a pulse signal that leads to powering the motor is enabled. The processor is further configured to determine that a speed associated with the motor is more than a threshold speed, wherein the speed associated with the motor is a modeled speed determined from a circuit model that represents the motor. The processor is further configured to determine whether a drive voltage output associated with the motor drive circuit is more than a threshold voltage. The processor is further configured to determine that a current applied to the motor is less than a threshold current. The processor is further configured to determine that the speed associated with the motor is zero. The processor is further configured to determine that at least one terminal of the motor is disconnected from the motor drive circuit in response to determining that the set of disconnection conditions is met for more than the threshold

time.

(10) Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 illustrates a diagram of an example HVAC system which includes motor-driven components;

(3) FIG. 2 illustrates a diagram of an example system configured to detect motor disconnection while a motor is switched off and while it is switched on;

(4) FIG. 3 is a diagram of an example motor drive circuit for use in the system of FIG. 2;

(5) FIG. 4 illustrates a plot of an example motor-drive signal provided by a transistor-pair stage of the example motor drive circuit of FIG. 3;

(6) FIG. 5 illustrates a flowchart of an example method for detecting motor disconnections when the motor is switched off, and

(7) FIG. 6 illustrates a flowchart of an example method for detecting motor disconnections when the motor is switched on.

DETAILED DESCRIPTION

(8) As described above, previous technologies fail to provide an efficient, secure, and reliable solution for detecting motor disconnections at terminals of motors, for example, motors used in Heating, Ventilation, and Air Conditioning (HVAC) systems). Embodiments of the present disclosure and its advantages may be understood by referring to FIGS. 1 through 6. FIGS. 1 through 6 are used to describe systems and methods for detecting motor disconnections at terminals of motors before the motor is switched on and while the motor is switched on.

(9) System Overview

(10) FIG. 1 illustrates a schematic diagram of an embodiment of an HVAC system **100**. The HVAC system **100** conditions air for delivery to a conditioned space. Various components of the HVAC system **100** may be motor-driven components including, but not limited to, the compressor **106**, the fan **110**, and the blower **128**, described in greater detail below. The conditioned space may be, for example, a room, a house, an office building, a warehouse, or the like. In some embodiments, the HVAC system **100** is a rooftop unit (RTU) that is positioned on the roof of a building and the conditioned air is delivered to the interior of the building. In other embodiments, portion(s) of the system may be located within the building and portion(s) outside the building. The HVAC system **100** may also include heating elements that are not shown here for convenience and clarity. The HVAC system **100** may be configured as shown in FIG. 1 or in any other suitable configuration. For example, the HVAC system **100** may include additional components or may omit one or more components shown in FIG. 1.

(11) The HVAC system **100** includes a working-fluid conduit subsystem **102**, a condensing unit **104**, an expansion valve **114**, a cooling unit **116**, a thermostat **134**, and a controller **138**. The working fluid conduit subsystem **102** facilitates the movement of a working fluid (e.g., a refrigerant) through a cooling cycle such that the working fluid flows as illustrated by the dashed arrows in FIG. 1. The working fluid may be any acceptable working fluid including, but not limited to, fluorocarbons (e.g., chlorofluorocarbons), ammonia, non-halogenated hydrocarbons (e.g., propane), hydrofluorocarbons (e.g., R-410A), or any other suitable type of refrigerant.

(12) The condensing unit **104** includes a compressor **106**, a condenser **108**, and a fan **110**. The compressor **106** is coupled to the working-fluid conduit subsystem **102** and compresses (i.e., increases the pressure of) the working fluid. The compressor **106** of condensing unit **104** may be a variable speed or multi-stage compressor. A variable speed compressor is generally configured to operate at different speeds to increase the pressure of the working fluid to keep the working fluid moving along the working-fluid conduit subsystem **102**. In the variable speed compressor configuration, the speed of compressor **106** can be modified to adjust the cooling capacity of the HVAC system **100**. Meanwhile, a multi-stage compressor may include multiple compressors, each configured to operate at a constant speed to increase the pressure of the working fluid to keep the working fluid moving along the working-fluid conduit subsystem **102**. In the multi-stage compressor configuration, one or more compressors can be turned on or off to adjust the cooling capacity of the HVAC system **100**.

(13) The compressor **106** is in signal communication with the controller **138** using a wired or wireless connection. The controller **138** provides commands or signals to control the operation of the compressor **106**. For example, the controller **138** may operate the compressor **106** in different modes corresponding to load conditions (e.g., the amount of cooling or heating required by the HVAC system **100**). As described in greater detail with respect to FIG. 2 below, the compressor **106** may be a motor-driven component. Accordingly, the controller **138** may provide a signal to a motor-drive circuit which powers a motor associated with the compressor **106**. The controller **138** is described in greater detail below with respect to FIG. 2.

(14) The condenser **108** is generally located downstream of the compressor **106** and is configured to remove heat from the working fluid. The fan **110** is configured to move air **112** across the condenser **108**. For example, the fan **110** may be configured to blow outside air through the condenser **108** to help cool the working fluid flowing therethrough. The fan **110** may be in signal communication with the controller **138** using a wired or wireless connection such that the controller **138** provides commands or signals to control the operation of the fan **110**. The fan **110** may be a motor-driven component. The controller **138** may provide a signal to a motor-drive circuit which powers a motor associated with the fan **110**. The cooled working fluid from the condenser **108** flows toward an expansion valve **114**.

(15) The expansion valve **114** is coupled to the working-fluid conduit subsystem **102** downstream of the condenser **108** and is configured to remove pressure from the working fluid. In this way, the working fluid is delivered to the cooling unit **116** and receives heat from airflow **118** to produce a conditioned airflow **120** that is delivered by a duct subsystem **122** to the conditioned space. In general, the expansion valve **114** may be a valve such as an expansion valve or a flow control valve (e.g., a thermostatic expansion valve (TXV) valve) or any other suitable valve for removing pressure from the working fluid while, optionally, providing control of the rate of flow of the working fluid. The expansion valve **114** may be in communication with the controller **138** (e.g., via wired and/or wireless communication) to receive control signals for opening and/or closing associated valves.

(16) The cooling unit **116** is generally any heat exchanger configured to provide heat transfer between air flowing through the cooling unit **116** (i.e., contacting an outer surface of one or more evaporator coils associated with the cooling unit **116**) and working fluid passing through the interior of the cooling unit **116**. The cooling unit **116** is fluidically connected to the compressor **106**, such that working fluid generally flows from the cooling unit **116** to the compressor **106**.

(17) A portion of the HVAC system **100** is configured to move air **118** across the cooling unit **116** and out of the duct sub-system **122** as conditioned air **120**. Return air **124**, which may be air returning from the building, fresh air from outside, or some combination, is pulled into a return duct **126**. A suction side of the blower **128** pulls the return air **124** through the duct **126**. The blower **128** discharges airflow **118** into a duct **130** from where the airflow **118** crosses the cooling unit **116** or heating elements (not shown) to produce the conditioned airflow **120**. The blower **128** is any

mechanism for providing a flow of air through the HVAC system **100**. For example, the blower **128** may be a constant-speed or variable-speed circulation blower or fan. Examples of a variable-speed blower include, but are not limited to, belt-drive blowers controlled by inverters, direct-drive blowers with electronic commuted motors (ECM), or any other suitable types of blowers.

(18) The blower **128** is in signal communication with the controller **138** using any suitable type of wired or wireless connection. The controller **138** is configured to provide commands or signals to the blower **128** to control its operation. The blower **128** may be a motor-driven component. The controller **138** may provide a signal to a motor-drive circuit which powers a motor associated with the blower **128**.

(19) The HVAC system **100** generally includes one or more sensors **132a,b** in signal communication with the controller **138**. The sensors **132a,b** may include any suitable type of sensor for measuring air temperature as well as other properties of a conditioned space (e.g., a room or building). The sensors **132a,b** may be positioned anywhere within the conditioned space, the HVAC system **100**, and/or the surrounding environment. For example, as shown in the illustrative example of FIG. **1**, the HVAC system **100** may include a sensor **132a** positioned and configured to measure a return air temperature (e.g., of airflow **124**) and a sensor **132b** positioned and configured to measure a supply or treated air temperature (e.g., of airflow **120**). In other examples, the HVAC system **100** may include sensors positioned and configured to measure any other suitable type of air temperature (e.g., the temperature of air at one or more locations within the conditioned space and/or an outdoor air temperature).

(20) The HVAC system **100** includes one or more thermostats **134**, for example, located within the conditioned space (e.g., a room or building). The thermostat **134** is generally in signal communication with the controller **138** using any suitable type of wired or wireless communications. The thermostat **134** may be a single-stage thermostat, a multi-stage thermostat, or any suitable type of thermostat as would be appreciated by one of ordinary skill in the art. The thermostat **134** is configured to allow a user to input a desired temperature via a temperature setpoint **136** for a designated space or zone such as a room in the conditioned space. The controller **138** may use information from the thermostat **134** such as the temperature setpoint **136** for controlling the compressor **106**, the fan **110**, and/or the blower **128**.

(21) As described above, in certain embodiments, connections between various components of the HVAC system **100** are wired. For example, conventional cable and contacts may be used to couple the controller **138** to the various components of the HVAC system **100**, including, the compressor **106**, the fan **110**, the expansion valve **114**, the blower **128**, sensor(s) **132**, and thermostat **134**. In some embodiments, a wireless connection is employed to provide at least some of the connections between components of the HVAC system **100**. In some embodiments, a data bus couples various components of the HVAC system **100** together such that data is communicated therebetween. In a typical embodiment, the data bus may include, for example, any combination of hardware, software embedded in a computer readable medium, or encoded logic incorporated in hardware or otherwise stored (e.g., firmware) to couple components of HVAC system **100** to each other. As an example and not by way of limitation, the data bus may include an Accelerated Graphics Port (AGP) or other graphics bus, a Controller Area Network (CAN) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCI-X) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or any other suitable bus or a combination of two or more of these. In various embodiments, the data bus may include any number, type, or configuration of data buses, where appropriate. In certain embodiments, one or more data buses (which may each include an address bus and a data bus) may couple the controller **138** to other components of the HVAC system **100**.

(22) Example Motor Disconnection Detection System

(23) FIG. 2 illustrates an example embodiment of a motor disconnection detection system **200**. The system **200** includes a motor-driven component **202**, a motor **204**, a motor drive circuit **206**, and a controller **220**. The motor-driven component **202** may be any of the components of the HVAC system **100**, including the compressor **106**, fan **110**, or blower **128** of FIG. 1. In other embodiments, the system **200** may include fewer components or additional components than those described above.

(24) System **200** is generally configured to detect a disconnection with respect to one or more terminals of the motor **204**. In some cases, a terminal (e.g., any of the terminals a, b, c) of the motor **204** may be disconnected due to hardware and/or software malfunctions. If the motor **204** is disconnected from the motor drive circuit **206** or the component **202**, the current approaches do not provide a solution to detect motor disconnection. In other words, using the current approaches, it is not known whether the motor is disconnected, much less whether the location of the disconnection is from a motor terminal or from a side of the motor **204**—from the front or the backend.

(25) System **200** is configured to provide a technical solution to the technical problem above and other technical problems. For example, the system **200** is configured to provide a method to detect the disconnection at one or more terminals of the motor **204**. The system **200** is configured to detect the motor disconnection while the motor **204** is switched off (e.g., before its operation starts). This operation is described in greater detail in FIGS. 2-5. The system **200** is further configured to detect the motor disconnection while the motor **204** is switched on. This operation is described in greater detail in FIGS. 2 and 6.

(26) System Components

(27) Example Motor

(28) Motor **204** is generally any appropriate motor configured to operate component **202**. For example, motor **204** may be a motor of a compressor (e.g., compressor **106** of FIG. 1), a fan (e.g., fan **110** of FIG. 1), or a blower (e.g., blower **128** of FIG. 1). The motor **204** may be any electrically powered motor that is powered by a motor-drive signal **212**. For example, the motor **204** may be an induction motor. Motor-drive signal **212** may be a pulse-width modulated signal, a DC voltage signal, and the like. Motor drive circuit **206** is generally any appropriate circuit for providing the motor-drive signal **212** in order to drive the motor **204**. For instance, the motor drive circuit **206** may be a direct current-alternating current (DC-AC) inverter circuit.

(29) Motor **204** may be configured to operate the component **202**. For example, the motor **204** may be configured to power the component **202** by communicating signals. The motor drive circuit **206** may receive control signals **210** from the controller **220** and send a motor drive signal **212** to the motor **204**. As described above, the motor-driven component **202** may be any of the compressor **106**, the fan **110**, or the blower **128** of the HVAC system **100**, described above. While the example of FIG. 2 is generally described in the context of operating along with or within an HVAC system, such as HVAC system **100**, it should be understood that system **200** may be used in any other appropriate context involving a motor-driven component **202** and is not limited to a motor-driven component **202** of an HVAC system.

(30) Example Motor Drive Circuit

(31) Motor drive circuit **206** is described in greater detail in FIG. 3. In brief, the motor drive circuit **206** is generally any circuit that is configured to power the motor **204**. The motor drive circuit **206** may include complementary transistors stages configured to drive DC signals to terminals of the motor **204** in response to being switched on or off according to the PWM control signals **210**. The motor drive circuit **206** may further include or be operably coupled to shunt resistors **208**. The shunt resistors **208** may act as current sensors configured to detect current received from different terminals of the motor **204**. The motor drive circuit **206** may be a pulse-width modulated power circuit where the modulated pulse signals **210** are communicated from the controller **220**.

(32) Example Controller

(33) Controller **220** may be the controller **138** of FIG. 1. The controller **220** may include a

processor **222**, a memory **226**, and an input/output (I/O) interface **224**. The memory **226** stores software instructions **228** that when executed by the processor **222** cause the processor **222** to perform one or more operations of the controller **220** described herein. In some embodiments, the controller **220** may include fewer components or additional components than those described above.

(34) The processor **222** includes one or more processors operably coupled to the memory **226**. The processor **222** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g. a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs) that communicatively couples to memory **226** and controls the operation of HVAC system **100** (see FIG. 1). The processor **222** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **222** is communicatively coupled to and in signal communication with the memory **226**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **222** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **222** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory **226** and executes them by directing the coordinated operations of the ALU, registers, and other components. The processor may include other hardware and software that operates to process information, control the HVAC system **100** (see FIG. 1), and perform any of the functions described herein (e.g., with respect to FIGS. 1-6). The processor **222** is not limited to a single processing device and may encompass multiple processing devices. Similarly, each of the controllers **220** is not limited to a single controller but may encompass multiple controllers.

(35) The memory **226** includes one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **226** may be volatile or non-volatile and may include read only memory (ROM), random access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **226** is operable to store software instructions **228** that when executed by the processor **222** cause the processor **222** to perform one or more operations of the controller **220** described herein. The memory **226** may further store control signals **210**, threshold values **230**, disconnection conditions **232**, and motor disconnection time counter **234**, sensor-less algorithm **236**, alert message **238**, and/or any data/instruction described in conjunction with FIGS. 1-6.

(36) The control signals **210** may include signals configured to instruct the motor drive circuit **206** to transmit motor drive signals **212** to the motor **204**. For example, the control signals **210** may include instructions in form of signals, modulated signals, pulse-width modulation signals, and the like.

(37) In one example, the control signals **210** may include signals to activate (e.g., turn on) one or more transistors in the motor drive circuit **206** and deactivate (e.g., turn off) one or more transistors in the motor circuit **206** so that a DC voltage signal is sent from the motor drive circuit **206** to a first terminal of the motor **204** and a returning DC voltage signal is received from second and third terminals of the motor **204**. This operation is described in greater detail in FIG. 3.

(38) The threshold values **230** may include any of the thresholds used to implement the functions described herein including thresholds associated with detecting a motor disconnection at one or more terminals. For example, the threshold values **230** may include a first current threshold associated with a first terminal of the motor **204**, a second current threshold associated with a second terminal of the motor **204**, and a third current threshold associated with a third terminal of the motor **204**. In some examples, the threshold values **230** may include voltage thresholds

associated with the motor **204** and/or component **202**, speed thresholds associated with the motor **204** and/or component **202**, and torque thresholds associated with any of the motor **204**, component **202**, a rotor (associated with the motor **204**), a stator (associated with the motor **204**), inverters (e.g., complementary stages associated with the motor **204** shown in FIG. 3), and/or other components of the motor **204**, component **202** and/or the HVAC system **100** of FIG. 1. Threshold values **230** may include thresholds associated with operation of the HVAC system **100** that indicate a motor disconnection at one or more terminals of the motor **204**. For example, threshold values may correspond to a threshold value or range of values associated with the speed or torque of motor **204** and/or the component **202**.

(39) The sensor-less algorithm **236** may be implemented by the processor **222** executing software instructions **228** and is generally configured to exert control signals to the circuit model of the motor **204** to estimate the modeled behavior of the motor **204**. The controller **220** may maintain a circuit model of the motor **204**, e.g., in software code or instructions **228**. The circuit model of the motor **204** may represent a real-time simulation of the motor **204**. The controller **220** may exert control signals to the circuit model of the motor **204** to estimate the modeled behavior of the motor **204**. This method may be referred to as a sensor-less algorithm **236** which observes motor parameters based on the circuit model of the motor **204**. For example, the sensor-less algorithm **236** may determine or estimate the current and voltage at each phase of the motor, flux, torque, speed, and angle of the rotor, among other parameters. In this manner, the controller **220** may implement the software code **228** (that includes the sensor-less algorithm **236**) to estimate the modeled speed associated with the motor **204**. In certain embodiments, the controller **220** may implement the software code **228** (that includes the sensor-less algorithm **236**) to estimate the modeled speed associated with the motor **204**.

(40) The I/O interface **224** is configured to communicate data and signals with other devices. For example, the I/O interface **224** may be configured to communicate electrical signals with components of the HVAC system **100** and system **200** including the compressor **106**, the fan **110**, the expansion valve **114**, the blower **128**, sensors **132a-d**, the thermostat **134**, and the motor drive circuit **206**. The I/O interface may receive, for example, thermostat calls, temperature setpoints, environmental conditions, sensor data, and an operating mode status for the HVAC system **100** and send electrical signals to the components of the HVAC system **100**. The I/O interface **224** may include ports or terminals for establishing signal communications between the controllers **220** and other devices. The I/O interface **224** may be configured to enable wired and/or wireless communications.

(41) One or more of the components of the controllers **220** described above may also or alternatively be included in the motor drive circuit **206** of FIG. 2. For example, the motor drive circuit **206** may include instances of the processor **222**, memory **226**, and/or I/O interface **224**. For example, if processor **222** is included in the motor drive circuit **206** of FIG. 2, the processor **222** of the motor drive circuit **206** may be configured to determine whether there is a disconnection at one or more terminals of the motor **204**, as described in greater detail below. If the memory **226** is included in the motor drive circuit **206** of FIG. 2, this memory **226** of the motor drive circuit **206** may store threshold values **230** (e.g., associated with whether there is a disconnection at one or more terminals of the motor **204**) among other data/instructions described herein.

(42) The motor drive circuit **206** may be connected to shunt resistors **208**. The shunt resistors **208** may act as current sensors for sensing current at the terminals of the motor **204** and/or current at inverters within the motor drive circuit **206**. In one example, the shunt resistors **208** may be current sensors (e.g., **132c, d**) described in FIG. 1. The motor drive circuit **206** may be connected to voltage sensors (e.g., **132c, d**) that are configured to sense voltage at the terminals of the motor **204**, similar to that described in FIG. 1.

(43) Operational Flow for Motor Disconnection while the Motor is Switched Off

(44) In operation, to determine the motor disconnection while the motor **204** is switched off, in

certain embodiments, the controller **220** may first determine whether the motor is powered up. To this end, the control **220** may determine whether the PWM signal **210** (e.g., PWM signal shown in FIG. **4**) is enabled. In normal operational conditions, the controller **220** may execute the software instructions **228** to communicate the control signals **210** (e.g. PWM signals) to the motor drive circuit **206**. In some cases, if there is a software or hardware malfunction at any of the components of the HVAC system **100** of FIG. **1** or any of the components of the system **200**, the controller **220** may disable the control signals **210** (e.g., not communicate the control signals **210** to the motor drive circuit **206**). In some cases, if an anomaly is detected with respect to any of the components **202**, the controller **220** may disable the communication of the control signals **210** to the motor drive circuit **206**. For example, if the controller **220** detects that a temperature of a component **202** is more than a threshold temperature (e.g., a threshold temperature value **230**), the controller **220** may disable the control signals **210** (e.g., communication of the control signals **210** may be disabled). In another example, if the controller **220** detects an unstable power supply (or any other anomaly), the communication of the control signals **210** may be disabled. In some examples the motor drive circuit **206** may disable the communication of the motor drive signals **212** to the motor **204** if the motor drive circuit **206** detects a software malfunction, a hardware malfunction, and/or an anomaly with respect to a component **202** or the motor **204**. By determining whether the PWM signal **210** is enabled, the controller **220** may exclude other cases that may lead to the current not flowing through the terminals of the motor **204**. In other words, in this way, false positive determinations of motor disconnections (with respect to disconnection at a motor terminal) are reduced or eliminated.

(45) Determining Whether the Alignment Procedure is being Performed

(46) In certain embodiments, the controller **220** may determine whether the motor **204** is in the alignment procedure. Before the alignment procedure, the position of the permanent magnet within the motor **204** is not known. However, the position of the windings within the motor **204** is known. Therefore, to align the permanent magnets to one of the windings, a DC voltage is injected (e.g., communicated) to a terminal (e.g., a phase) of the motor **204**. In response, current flows through the winding associated with the terminals, and an electromagnet field is established along the phases of the motor **204**. In response, the permanent magnet will position itself to be aligned with the established electromagnet field induced along the phase of the motor **204**.

(47) In certain embodiments, before starting field magnet control with respect to permanent magnets (not shown) within the motor **204**, it is desired to align the permanent magnets with at least one of the windings within the motor **204**. When the permanent magnet is aligned with at least one of the windings, that position may be taken as the reference position of the rotor angle being zero. From that point on, the angle of the rotor may be calculated with respect to the reference position. In this manner, the angle of the rotor may be calculated while the rotor is rotating.

(48) To initiate the alignment procedure, the controller **220** may communicate, to the motor drive circuit **206**, an alignment control signal that indicates to inject a DC signal to the motor **204**. The alignment control signal may be included in the control signals **210**. In response, the motor drive circuit **206** may communicate the motor drive signals **212** that includes DC signals to the phases a, b, and c of the motor **204** to induce the current through the phases and align the rotor angle with a known angle. The aligned rotor angle is used for motor control purposes.

(49) The motor drive circuit **206** may inject the DC signals when triggered by the software control instructions **228** from the controller **220**. The software control instructions **228** may control the PWM control signals **210** which then triggers the motor drive circuit **206** to inject the DC signals to phases a, b, and c of the motor **204**. The amplitudes of DC signals are controlled by the software control instructions **228**. In one example, the same amplitude DC voltage signal may be applied to the phases a, b, and c. The control signals **210** may be used to perform the alignment procedure described above.

(50) Detecting Motor Disconnection Based on Evaluating the Current at Each Terminal

(51) Referring to FIG. 2, while the alignment procedure is being performed, the controller 220 may evaluate the current signal values with respect to a respective expected current value at each terminal (e.g., each of the terminals a, b, and c) of the motor 204. For example, the controller 220 may communicate the control signal 210 that indicates to transmit a DC voltage signal to one or more terminals of the motor 204 in response to determining that the permanent magnet associated with the motor 204 is being aligned with a winding of the motor 204. In response, the controller 220 may determine the current signal values at the terminals of the motor 204.

(52) The controller 220 may compare each of the determined current signals with a respective threshold current value (e.g., threshold current value 230). Each respective threshold current value may be related to an expected current value at a respective terminal. For example, if a first expected current value at a first terminal is 4 milli-ampere (mA), the first threshold current value may be the 4 mA or within a range of 4 mA. In another example, if a second expected current value at a second terminal is 2 mA, the second threshold current value may be 2 mA or within a range of 2 mA. The expected current values may be based on current values determined during normal conditions where there is no disconnection at a terminal of the motor 204.

(53) If the controller 220 determines that a first determine current value (determined at a first terminal a) is less than a respective current value, it may be determined that the first terminal a of the motor is disconnected, e.g., from the motor drive circuit 206. Otherwise, it may be determined that the first terminal a is connected to the motor drive circuit 206. The controller 220 may perform similar operations for other current values at other terminals. For example, the controller 220 may determine a second current value at a second terminal b of the motor 204. The controller 220 may then compare the second current value with a second threshold current value. The second current value may be different from one or more other current values. The controller 220 may determine whether the second current value is more than the second threshold current value. If it is determined that the second current value is more than the second threshold current value, the controller 220 may determine that the second terminal b of the motor 204 is disconnected, e.g., from motor drive circuit 206. Otherwise, it may be determined that the second terminal is connected to the motor drive circuit 206. Similar operations may be performed for terminal c of the motor 204.

(54) FIG. 3 illustrates a circuit diagram of an example of the motor drive circuit 206 of FIG. 2. As shown in FIG. 3, motor drive circuit 206 may include three stages 302a-c, where each stage 302a-c includes a corresponding pair of transistors (transistors 304a-c and 306a-c). Each stage 302a-c may be referred to as an inverter. For example, the first stage 302a has a first transistor 304a and a second transistor 306a. Similarly, stages 302b and 302c have corresponding first transistors 304b and 304b, respectively, and second transistors 306b and 306c, respectively. Signal(s) and/or commands (e.g., control signal(s) 210) received from the controller 220 (see FIG. 2) provides instructions for the operation of each stage 302a-c. For example, the controller 220 may provide signal(s) 210 and/or command(s) associated with the frequency at which transistors 304a-c and 306a-c are switched on and off (e.g., to provide an appropriately modulated signal for powering the motor 204). Each stage 302a-c has a corresponding signal output 308a-c. The motor-drive signal 212 of FIG. 2 may include signals from each of the outputs 308a-c. An example of signals 308a-c is shown in FIG. 4.

(55) In FIG. 3, some of the internal components of three phases of the motor 204 are illustrated. Each phase of the motor 204 may include a resistor (R) and an inductor (L). The RL model of a phase may be used to determine the current and voltage at each phase when the motor is switched off. When the motor 204 is switched on, other components may be included in modeling each phase, such as a voltage source. In certain embodiments, the motor 204 may include some, all, or additional components with respect to those shown in FIG. 3. The RL circuit model is used for illustration purposes and is not meant to limit the scope of the present disclosure.

(56) During operation of the motor drive circuit 206, when a first transistor 304a-c of a given stage

302a-c is switched on, a second transistor **306a-c** is switched off (and vice versa). For example, if, at a given time during operation of stage **302a**, transistor **304a** is in an “on state” (e.g. corresponding to a state in which current can flow through transistor **304a**), the associated second transistor **306a** is in an off state (e.g. corresponding to a state in which current cannot flow through transistor **306a**).

(57) In an example operation of system **200**, the controller **220** provides (e.g., transmits) a control signal **210** to the motor drive circuit **206**. The control signal **210** causes stages **302a-c** to operate such that a switching frequency of each stage (i.e., the frequency at which each of signals **308a-c** switch between an on and off state, which is associated with the frequency at which transistors **304a-c** and **306a-c** are switched on and off). For example, the control signal **210** may indicate to transmit a DC voltage signal to the phase or terminal a of the motor **204**. In this example, the transistors **304a**, **306b**, and **306c** are switched on and other transistors are switched off. This leads to the current ($I_{sub.1}$) flowing from the transistor **304a** to the terminal a. The current $I_{sub.1}$ is divided and flows to terminals b and c. The current $I_{sub.1}$ is divided depending on the equivalent impedance along each phase/leg b and c. For example, if phases b and c have an equal impedance compared to each other, the current $I_{sub.1}$ is divided into two equal parts, $I_{sub.1}/2$ flows to terminal b and $I_{sub.1}/2$ flows to terminal c. The current $I_{sub.2}$ flows out of terminal b, and current $I_{sub.3}$ flows out of terminal c. The current $I_{sub.2}$ flows through the switched-on transistor **306b**, and current $I_{sub.3}$ flows through the switched-on transistor **306c**. The currents $I_{sub.2,3}$ are flown back to the motor drive circuit **206** and the controller **220**.

(58) The controller **220** may measure the currents $I_{sub.1,2,3}$, and compares each of them to a respective threshold current level to determine whether a current $I_{sub.1,2,3}$ is less than a respective threshold current level, similar to that described in FIG. 2. If it is determined that any of the currents $I_{sub.1,2,3}$ is less than a respective threshold current, it may be determined that a respective terminal is disconnected.

(59) Operational Flow for Motor Disconnection while the Motor is Switched on

(60) Referring back to FIG. 2, the system **200** may further be configured to detect motor disconnection while the motor **204** is switched on (i.e., running). When the motor **204** is switched on, certain motor parameters are not constant and vary depending on the operation of the motor **204**, the current applied to the motor **204**, among other factors. For example, when the motor **204** is switched on, the circuit model of the motor **204** may include a voltage source that varies as a function of the current applied to the motor **204**, the inductance of the motor **204** may vary as a function of the current applied to the motor **204**, among others. Thus, in situations where the motor **204** is switched on, relying on static RL model may not be reliable. Furthermore, in such situations, a DC voltage signal may not be reliable for motor disconnection detection. Therefore, the operational flow of system **200** for motor disconnection while the motor **204** is switched on may rely on various factors, such as, the control signals **210** (e.g., PWM signals) are enabled, a modeled speed of the motor **204**, the drive voltage output (e.g., motor drive signal **212**) that is provided to the motor **204** from inverter stages within the motor drive circuit **206**, the current applied to the motor **204**, and estimated impedance associated with the motor **204** that is determined from the drive voltage output and the current applied to the motor **204**. These factors may be referred to as a set of motor disconnection conditions **232**. The following description describes how the system **200** determines whether each of the set of motor disconnection conditions **232** is evaluated to detect motor disconnection while the motor **204** is switched on.

(61) Determining Whether a Set of Motor Disconnection Conditions is Met

(62) In operation, the operational flow of the system **200** for motor disconnection while the motor **204** is switched on may begin when the controller **220** evaluates each of the set of motor disconnection conditions **232**. The controller **220** may determine whether the control signals **210** are enabled, similar to that described above with respect to the operational flow of the system **200** for motor disconnection while the motor **204** is switched off. In normal operational condition, the

controller **220** may execute the software instructions **228** to communicate the control signals **210** to the motor drive circuit **206**. In response, the motor drive circuit **206** may generate and communicate the motor drive signals **212** to the motor **204**. However, if there is a software malfunction, a hardware malfunction, and/or an anomaly detected with respect to any of the components of the systems **100**, **200**, the control signals **210** may be disabled (e.g., not communicated to the motor drive circuit **206**). Thus, in this manner, the controller **220** may determine whether the motor **204** is receiving signals by determining whether the control signals **210** are enabled.

(63) The controller **220** may maintain a circuit model of the motor **204**, e.g., in software code or instructions **228**. The controller **220** may use the sensor-less algorithm **236** to estimate the modeled speed associated with the motor **204**. The controller **220** may determine whether the modeled speed (associated with the motor **204** is more than a threshold speed, e.g., in response to determining that the control signals **210** are enabled. The modeled speed associated with the motor **204** may be the speed at which the rotor of the motor **204** rotates along its central axis. The threshold speed may be one of the threshold values **230**. For example, the threshold speed may be 100 round-per-minute (RPM), 200 RPM, and the like.

(64) The controller **220** may determine whether a drive voltage output associated with the motor drive circuit **206** that is provided to the motor **204** is more than a threshold voltage, e.g., in response to determining that the modeled speed associated with the motor **204** is more than the threshold speed. The drive voltage output may be determined by a voltage sensor (e.g., a sensor **132c, d** described in FIG. **1**). The drive voltage output may be the motor drive signal **212** that is provided from the inverters stages within the motor drive circuit **206** to the motor **204** as shown in FIG. **3**. The same or similar drive voltage output may also be applied to the circuit model of the motor **204**. The threshold voltage may be one of the threshold values **230**. For example, the threshold voltage may be 110 voltage (v), and the like.

(65) The controller **220** may determine whether the current applied to the motor **204** is less than a threshold current, e.g., in response to determining that the drive voltage output that is provided to the motor **204** is more than the threshold voltage. The current applied to the motor **204** may be determined by a current sensor, such as a shunt resistor **208** (R) which may be an instance of a sensor **132c,d** described in FIG. **1**. The threshold current may be one of the threshold values **230**. For example, the threshold current may be 0.03 A, 0.02 A, and the like.

(66) The circuit model of the motor **204** may not be configured to detect the motor disconnections. For example, in the case of a motor disconnection where little or no current flows through the motor **204**, it is practically impossible for the rotor to rotate at a high speed (e.g., the estimated speed more than the threshold speed as described above). However, the circuit model of the motor **204** may indicate that the modeled speed of the rotor is high (e.g., more than the threshold speed). Therefore, this operational flow of the system **200** may be configured to detect motor disconnection scenarios that are not caught by current circuit models of the motor **204**.

(67) The controller **220** may determine whether the speed associated with the motor **204** is zero, e.g., in response to determining that the current applied to the motor **204** is less than the threshold current. In some cases, the motor **204** may already be in motion, for example due to manual forces applied to the motor **204**, e.g., by an operator. Thus, to exclude such cases from motor disconnection detection, the controller **220** may determine whether the speed associated with the motor **204** is zero. If the speed associated with the motor **204** is not zero, the issues with the motor **204** (if any) may have been influenced by other factors other than solely the operations and connections of the motor **204**.

(68) In this manner, the controller **220** may determine whether the set of disconnection conditions **232** is met, where the set of disconnection conditions **232** may include the pulse signal that leads to powering the motor **204** is enabled, the speed associated with the motor **204** is more than the threshold speed, the drive voltage output provided to the motor **204** is more than the threshold

voltage, the current applied to the motor **204** is less than the threshold current, and the speed associated with the motor **204** is zero. Although certain embodiments and examples described above indicate evaluating the set of disconnection conditions **232** in a specific order, the controller **220** may evaluate the set of disconnection conditions **232** in any suitable order.

(69) The controller **220** may determine whether the set of disconnection conditions **232** is met for more than a threshold time. The threshold time may be one of the threshold values **230**. For example, the threshold values may be ten seconds, twenty seconds, and the like. The evaluation time while the set of disconnection conditions **232** may be evaluated may be tracked by a motor disconnection time counter **234**. If the controller **220** determines that the speed associated with the motor **204** is zero, the controller **220** may determine whether the motor disconnection time counter **234** has passed the threshold time. If it is determined that the motor disconnection time counter **234** has passed the threshold time, the controller **220** may determine that the motor **204** is disconnected (at least from one terminal) from the motor drive circuit **206**. In other words, if the controller **220** determines that the set of disconnection conditions **232** is met for more than the threshold time, the controller **220** may determine that there is a disconnection at the motor **204**. In response, the controller **220** may set the corresponding fault handler flag (e.g., included in the software instructions **228**) to true. The controller **220** may also set the motor disconnection time counter **234** to zero. If, however, it is determined that the motor disconnection time counter **234** has not reached the threshold time yet, the controller **220** may increase the motor disconnection time counter **234** by a unit time, e.g., 0.01 second and the like. If the controller **220** determines that the speed associated with the motor **204** is more than zero, the controller **220** may evaluate the motor disconnection time counter **234** and determine whether the motor disconnection time counter **234** is more than zero. For example, if it is determined that the motor disconnection time counter **234** is more than zero, the controller **220** may decrease the motor disconnection time counter **234** by a unit time, e.g., 0.01 second and the like.

(70) In certain embodiments, the controller **220** may determine the impedance (z) associated with the motor **204** by dividing the determine the drive voltage output by the determined current applied to the motor **204**. In certain embodiments, it may be determined that the motor is disconnected if the determined impedance associated with the motor **204** is more than a threshold impedance in combination with determining that the estimated speed associated with the rotor (determined by the sensor-less algorithm **236**) is more than the threshold speed.

(71) Example PWM Signal

(72) FIG. 4 shows an example of a voltage signal **402** and current signal **404** provided by one of the stages **302a-c** of the motor drive circuit **206** (e.g., as signals **308a-c** of FIG. 3). The voltage signal **402** and current signal **404** of FIG. 4 are at an initial switching frequency associated with the normal operation of the component (e.g., motor-driven component **202** of FIG. 2). During the “off” times shown in FIG. 4, the voltage signal **402** is relatively large and the current signal **404** is negligible or zero. During the “on” times shown in FIG. 4, the voltage signal **402** is at a relatively small but non-zero value and the current signal **404** is relatively large.

(73) The switching frequency for a signal **308a-c** of a given stage **302a-c** corresponds to the rate at which the corresponding transistors **304a-c** and **306a-c** are switched on and off (e.g., the number of times the current signal **404** switches between on and off states per unit of time). The switching frequency is inversely proportional to the period **406** shown in FIG. 4. For example, the initial switching frequency during operation of system **200** maybe 16 kHz. A transition period **408** occurs between the off and on times. Power losses, which are proportional to the current signal **404** multiplied by the voltage signal **402**, are illustrated by the shaded regions in FIG. 4. Power losses during the “on” periods are referred to as conduction losses. Power losses during each switching transition period **408** are referred to as switching losses.

(74) Example Method for Detecting Motor Disconnection while the Motor is Switched Off

(75) FIG. 5 illustrates an example method **500** of system **200** of FIG. 2 for detecting motor

disconnection while the motor **204** is switched off. Modifications, additions, or omissions may be made to method **500**. Method **500** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the system **100**, system **200**, the controller **220**, or components of any of thereof performing operations, any suitable system or components of the system may perform one or more operations of the method **500**. For example, one or more operations of method **500** may be implemented, at least in part, in the form of software instructions **228** of FIG. 2, stored on tangible non-transitory computer-readable media (e.g., memory **226** of FIG. 2) that when run by one or more processors (e.g., processor **222** of FIG. 2) may cause the one or more processors to perform operations **502-518**.

(76) At operation **502**, the controller **220** determines whether the control signal **210** is enabled. The control signal **210** may include PWM signals (see FIG. 4), pulse control signals, and the like. The control signal **210** may be used to control the motor drive circuit **206**, similar to that described in FIGS. 2 and 3. If it is determined that the control signal **210** is enabled, method **500** proceeds to operation **504**. Otherwise, method **500** may end.

(77) At operation **504**, the controller **220** determines whether the motor **204** is in the alignment procedure, according to certain embodiments described herein. In this process, the controller **220** may determine whether the permanent magnet associated with the motor **204** is being aligned with a winding associated with the motor **204**, similar to that described in FIG. 2. The control signal **210** may cause the motor drive circuit **206** to inject DC signals to the terminals of the motor **204**. The DC signals may induce current along the phases of the motor **204** which leads to aligning the permanent magnet of the motor **204** with the phase of the motor **204**, similar to that described in FIG. 2. If it is determined that the motor **204** is in alignment procedure, method **500** proceeds to operation **506**. Otherwise, method **500** may end.

(78) At operation **506**, the controller **220** communicates, to a motor drive circuit **206**, a control signal **210** that indicates to transmit a DC signal to the motor **204**. For example, the control signal **210** may be the control signal that is used for the alignment procedure or an additional control signal.

(79) At operation **508**, the controller **220** selects a phase of the motor **204**. The phase of the motor **204** may also be referred to as a terminal or a leg of the motor **204**. The controller **220** may iteratively select a phase of the motor **204** until all the phases of the motor **204** are evaluated. In case, the motor **204** is a three-phase motor, each of the three phases of the motor is iteratively selected.

(80) At operation **510**, the controller **220** determine a current signal value at the selected phase/terminal/leg of the motor **204**. For example, the controller **220** may determine the I.sub.1 if the phase a is selected, I.sub.2 if the phase b is selected, and I.sub.3 if the phase c is selected. The controller **220** may determine the current signal value by using a current sensor, a shunt resistor **208**, and the like, similar to that described in FIG. 2.

(81) At operation **512**, the controller **220** compares the determined current signal value with a respective threshold current value. The respective threshold current value may be different from other threshold current values, similar to that described in FIGS. 2 and 3. The respective threshold current value may be one of the threshold values **230**.

(82) At operation **514**, the controller **220** determines whether the current signal value is less than the respective threshold current value. If it is determined that the current signal value is less than the respective threshold current value, method **500** may proceed to operation **516**. Otherwise, method **500** may end.

(83) At operation **516**, the controller **220** determines that the motor terminal where the current signal value is determined to be less than the threshold current value is disconnected. For example, the controller **220** may determine that the motor terminal is disconnected from the motor drive circuit **206**. In another example, the controller **220** may determine that the motor terminal is

disconnected from the component **202**. In response to detecting the motor disconnection, the controller **220** may communicate an alert message **238** that indicates that the particular terminal of the motor is disconnected from the motor drive circuit **206**. For example, the controller **220** may communicate the alert message **238** to a computing device associated with an operator and/or display the alert message **238** on a display screen associated with the controller **220**. In certain embodiments, the controller **220** may determine that the motor terminal is disconnected, if the determined current signal value is determined to be less than the threshold current value for more than a threshold time window (e.g., more than ten seconds, twenty seconds, etc.), for example, while the control signal **210** is enabled and the alignment procedure is taking place.

(84) At operation **518**, the controller **220** may determine whether to select another phase of the motor **204**. The controller **220** may determine to select another phase of the motor **204** if at least one phase is left for evaluation. If it is determined that at least one phase is left for evaluation, method **500** may return to operation **508**. Otherwise, method **500** may end. The controller **220** may perform the method **500** periodically, for example, every 0.01 seconds, 0.02 seconds, and the like. In certain embodiments, the motor drive circuit **206** may transmit the DC signal to at least one of the three terminals of the motor **204** in response to receiving the control signal **210** from the controller **220**. For example, the motor drive circuit **206** may transmit the DC voltage signal to the terminal a, similar to that described in FIG. 3.

(85) In certain embodiments, the motor drive circuit **206** may transmit the DC voltage signal to the first terminal (e.g., terminal a) of the motor **204** when a first transistor **304a** is switched on and the second transistor **306a** is switched off, similar to that described in FIG. 3. In such embodiments, the motor drive circuit **206** may receive a second DC voltage signal from a second terminal (e.g., terminal b) of the motor **204** when a third transistor **306b** is switched on and a fourth transistor **304b** is switched off. Similarly, the motor drive circuit **206** may receive a third DC voltage signal from a third terminal (e.g., terminal c) of the motor **204** when a fifth transistor **306c** is switched on and a sixth transistor **304b** is switched off.

(86) Example Method for Detecting Motor Disconnection while the Motor is Switched on

(87) FIG. 6 illustrates an example method **600** of system **200** of FIG. 2 for detecting motor disconnection while the motor **204** is switched on. Modifications, additions, or omissions may be made to method **600**. Method **600** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the system **100**, system **200**, the controller **220**, or components of any of thereof performing operations, any suitable system or components of the system may perform one or more operations of the method **600**. For example, one or more operations of method **600** may be implemented, at least in part, in the form of software instructions **228** of FIG. 2, stored on non-transitory, tangible, machine-readable media (e.g., memory **226** of FIG. 2) that when run by one or more processors (e.g., processor **222** of FIG. 2) may cause the one or more processors to perform operations **602-622**.

(88) At operation **602**, the controller **220** determines whether the control signal **210** is enabled. For example, the controller **220** may determine that the control signal **210** is enabled if there is no anomaly, hardware malfunction, software malfunction, or the like detected, similar to that described in FIG. 2. If it is determined that the control signal **210** is enabled, method **600** may proceed to operation **604**. Otherwise, method **600** may proceed to operation **620**.

(89) At operation **604**, the controller **220** determines whether the modeled speed associated with the motor **204** is more than the threshold speed. The controller **220** may determine the modeled speed of the rotor associated with the motor **204** and compare it to the threshold speed, similar to that described in FIG. 2. If it is determined that the modeled speed associated with the motor **204** is more than the threshold speed, method **600** may proceed to operation **606**. Otherwise, the method **600** may proceed to operation **620**.

(90) At operation **606**, the controller **220** determines whether the drive voltage output is more than

the threshold voltage. For example, the controller **220** may determine the drive voltage output by a voltage sensor at the motor **204** and compare it to the threshold voltage, similar to that described in FIG. **2**. If it is determined that the drive voltage output is more than the threshold voltage, method **600** may proceed to operation **608**. Otherwise, method **600** may proceed to operation **620**.

(91) At operation **608**, the controller **220** determines whether the current applied to the motor **204** is less than a threshold current. For example, the controller **220** may determine the current applied to the motor **204** by a shunt resistor **208** and compare it to the threshold current, similar to that described in FIG. **2**. If it is determined that the applied current is less than the threshold current, method **600** proceeds to operation **610**. Otherwise, method **600** proceeds to operation **620**.

(92) At operation **610**, the controller **220** determines whether the speed associated with the motor **204** is zero. If it is determined that the speed associated with the motor **204** is more than zero, it may mean that the motor **204** is rotating by external or manual forces, for example, by an operator. In such cases, controller **220** may apply an opposite force to bring the motor **204** to a stop position. If it is determined that the speed associated with the motor **204** is zero, the method **600** may proceed to operation **612**. Otherwise, method **600** may proceed to operation **620**. The operations **602** to **610** may correspond to operations where the disconnection conditions **232** are evaluated.

(93) At operation **612**, the controller **220** determines whether the motor disconnection time counter **234** has reached the threshold time. In other words, the controller **220** determines whether the set of motor disconnection conditions **232** is met for more than the threshold time. If it is determined that the motor disconnection time counter **234** has reached the threshold time, the method **600** proceeds to operation **614**. Otherwise, the method **600** may proceed to operation **618**.

(94) At operation **614**, the controller **220** determines that the motor **204** is disconnected from the motor drive circuit **206**. The controller **220** may also determine a particular motor terminal (e.g., terminal a, b, c) that is disconnected. In response to detecting the motor disconnection, the controller **220** may communicate an alert message **238** that indicates at least one of the terminals of the motor is disconnected from the motor drive circuit **206**. For example, the controller **220** may communicate the alert message **238** to a computing device associated with an operator and/or display the alert message **238** on a display screen associated with the controller **220**.

(95) At operation **616**, the controller **220** sets the motor disconnection time counter **234** to zero. At operation **618**, the controller **220** increases the motor disconnection time counter **234** by a unit time, similar to that described in FIG. **2**.

(96) At operation **620**, the controller **220** determines whether the motor disconnection time counter **234** is more than zero. If it is determined that the motor disconnection time counter **234** is more than zero, method **600** proceeds to operation **622**. Otherwise, method **600** ends.

(97) At operation **622**, the controller **220** decreases the motor disconnection time counter **234** by a unit time, similar to that described in FIG. **2**. In certain embodiments, the method **600** may be performed periodically, for example, every 0.1 seconds, 0.2 seconds, and the like.

(98) In certain embodiments, the motor disconnection time counter **234** may be triggered and started when the controller **220** initiates evaluating the set of disconnection conditions **232**. In certain embodiments, determining that the motor **204** is disconnected from the motor drive circuit **206** may be in response to determining that the motor disconnection time counter **234** indicates that the threshold time is reached.

(99) In certain embodiments, if the controller **220** determines that any of the disconnection conditions **232** is not met, the controller **220** may determine whether the motor disconnection time counter **234** is more than zero seconds. In response to determining that the motor disconnection time counter **234** is more than zero seconds, the controller **220** may decrement the motor disconnection time counter **234** by a unit time. In certain embodiments, in response to determining that the set of disconnection conditions **232** is met and the motor disconnection time counter **234** has not reached the threshold time, the controller **220** may increment the motor disconnection time counter **234** by a unit time. In certain embodiments, in response to determining that the set of

disconnection conditions 232 is met for more than the threshold time, the controller 220 may reset the motor disconnection time counter 234.

(100) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

(101) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(102) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system for detecting motor disconnection while a motor is switched on, comprising: a motor configured to operate a motor-driven component, wherein the motor is associated with a motor drive circuit that supplies power to the motor; and a processor communicatively coupled to the motor drive circuit, and configured to: determine whether a set of disconnection conditions is met for more than a threshold time, comprising: determine that a pulse signal that leads to powering the motor is enabled; determine that a modeled speed associated with the motor is more than a threshold speed, wherein the modeled speed determined is from a circuit model that represents the motor; determine that a drive voltage output associated with the motor drive circuit is more than a threshold voltage; determine that a current applied to the motor is less than a threshold current; and determine that a speed associated with the motor is zero; and in response to determining that the set of disconnection conditions is met for more than the threshold time, determine that at least one terminal of the motor is disconnected from the motor drive circuit.

2. The system of claim 1, wherein: the processor is further communicatively coupled to a memory configured to store a motor disconnection time counter that is set to zero, wherein the motor disconnection time counter is started when the processor initiates determining whether the set of disconnection conditions is met; and determining that at least one of terminals of the motor is disconnected from the motor drive circuit is further in response to determining that the motor disconnection time counter indicates that the threshold time is reached.

3. The system of claim 2, wherein the processor is further configured to: in response to determining that the any one or more of the set of disconnection conditions is not met, determine whether the motor disconnection time counter is more than zero seconds; and in response to determining that the motor disconnection time counter is more than zero seconds, decrease the motor disconnection time counter by a unit time.

4. The system of claim 2, wherein the processor is further configured to, in response to determining that the set of disconnection conditions is met and the motor disconnection time counter has not reached the threshold time, increase the motor disconnection time counter by a unit time.

5. The system of claim 2, wherein the processor is further configured to, in response to determining

that the set of disconnection conditions is met for more than the threshold time, reset the motor disconnection time counter to zero.

6. The system of claim 1, wherein the set of disconnection conditions comprises: the pulse signal that leads to powering the motor is enabled; the modeled speed associated with the motor is more than the threshold speed; the drive voltage output associated with the motor drive circuit is more than the threshold voltage; the current at the motor is less than the threshold current; and the speed associated with the motor is zero.

7. The system of claim 1, wherein: the processor is further configured to determine an impedance associated with the motor by dividing the drive voltage output by the current; and determining that at least one of terminals of the motor is disconnected from the motor drive circuit is further in response to determining that the impedance is more than a threshold impedance.

8. A method for detecting motor disconnection while a motor is switched on, comprising: determining whether a set of disconnection conditions is met for more than a threshold time, comprising: determining that a pulse signal that leads to powering the motor is enabled, wherein the motor is associated with a motor drive circuit that supplies power to the motor; determining that a modeled speed associated with the motor is more than a threshold speed, wherein the modeled speed is determined from a circuit model that represents the motor; determining that a drive voltage output associated with the motor drive circuit is more than a threshold voltage; determining that a current applied to the motor is less than a threshold current; and determining that a speed associated with the motor is zero; and in response to determining that the set of disconnection conditions is met for more than the threshold time, determining that at least one terminal of the motor is disconnected from the motor drive circuit.

9. The method of claim 8, wherein: the method further comprises storing a motor disconnection time counter that is set to zero, wherein the motor disconnection time counter is started when evaluating the set of disconnection conditions is initiated; and determining that at least one of terminals of the motor is disconnected from the motor drive circuit is further in response to determining that the motor disconnection time counter indicates that the threshold time is reached.

10. The method of claim 9, further comprising: in response to determining that the any one or more of the set of disconnection conditions is not met, determining whether the motor disconnection time counter is more than zero seconds; and in response to determining that the motor disconnection time counter is more than zero seconds, decreasing the motor disconnection time counter by a unit time.

11. The method of claim 9, further comprising, in response to determining that the set of disconnection conditions is met and the motor disconnection time counter has not reached the threshold time, increasing the motor disconnection time counter by a unit time.

12. The method of claim 9, further comprising, in response to determining that the set of disconnection conditions is met for more than the threshold time, resetting the motor disconnection time counter to zero.

13. The method of claim 8, wherein the set of disconnection conditions comprises: the pulse signal that leads to powering the motor is enabled; the modeled speed associated with the motor is more than the threshold speed; the drive voltage output associated with the motor drive circuit is more than the threshold voltage; the current at the motor is less than the threshold current; and the speed associated with the motor is zero.

14. The method of claim 8, wherein: to the method further comprises determining an impedance associated with the motor by dividing the drive voltage output by the current; and determining that at least one of terminals of the motor is disconnected from the motor drive circuit is further in response to determining that the impedance is more than a threshold impedance.

15. A non-transitory computer-readable medium storing instructions that when executed by a processor, cause the processor to: determine whether a set of disconnection conditions is met for more than a threshold time, comprising: determine that a pulse signal that leads to powering a

motor is enabled, wherein the motor is associated with a motor drive circuit that supplies power to the motor; determine that a modeled speed associated with the motor is more than a threshold speed, wherein the modeled speed is determined from a circuit model that represents the motor; determine that a drive voltage output associated with the motor drive circuit is more than a threshold voltage; determine that a current applied to the motor is less than a threshold current; and determine that a speed associated with the motor is zero; and in response to determining that the set of disconnection conditions is met for more than the threshold time, determine that at least one terminal of the motor is disconnected from the motor drive circuit.

16. The non-transitory computer-readable medium of claim 15, wherein: the instructions further cause the processor to store a motor disconnection time counter that is set to zero, wherein the motor disconnection time counter is started when the processor initiates determining whether the set of disconnection conditions is met; and determining that at least one of terminals of the motor is disconnected from the motor drive circuit is further in response to determining that the motor disconnection time counter indicates that the threshold time is reached.

17. The non-transitory computer-readable medium of claim 16, wherein the instructions further cause the processor to: in response to determining that the any one or more of the set of disconnection conditions is not met, determine whether the motor disconnection time counter is more than zero seconds; and in response to determining that the motor disconnection time counter is more than zero seconds, decrease the motor disconnection time counter by a unit time.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to communicate an alert message that indicates at least one of terminals of the motor is disconnected from the motor drive circuit in response to determining that the at least one of terminals of the motor is disconnected from the motor drive circuit.

19. The non-transitory computer-readable medium of claim 15, wherein the motor powers a motor-driven component that is a fan, a compressor, or a blower of a heating, ventilation, and air conditioning (HVAC) system.

20. The non-transitory computer-readable medium of claim 15, wherein the set of disconnection conditions comprises: the pulse signal that leads to powering the motor is enabled; the modeled speed associated with the motor is more than the threshold speed; the drive voltage output associated with the motor drive circuit is more than the threshold voltage; the current at the motor is less than the threshold current; and the speed associated with the motor is zero.
